

Multiple Choice (10 marks)

1. What is the output of $4*1**4$ in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16
2. Which of the following sets of bit strings CANNOT be described with a regular expression?
 - (a) All bit strings whose number of zeros is a multiple of five
 - (b) All bit strings starting with a zero and ending with a one
 - (c) All bit strings with an even number of zeros
 - (d) All bit strings with more ones than zeros
3. In python 3, which of the following is floor division?
 - (a) /
 - (b) //
 - (c) %
 - (d) |
4. Let $l = [1,2,3,4]$. What is $\max(l)$ in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x**2]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100

True / False (10 marks)

Answer True or False

6. True or False: In a complete K -ary tree of depth N , the ratio of nonterminal nodes to total nodes is approximately $1/K$.
7. True or False: $O(\log n)$ is smaller than $O(1)$ since logarithmic growth decreases faster than constant growth.
8. True or False: A_3 -way, set-associative cache is effective only when 3 or fewer processes are running on the processor.
9. True or False: The function $O(\log \log N)$ grows slower than $O(\log N)$, making it one of the slowest growing functions.
10. True or False: The maximum possible degree of the minimal-degree interpolating polynomial $p(x)$ for $n + 1$ distinct points is n .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to search a literals string in a string and also find the location within the original string where the pattern occurs.
12. Write a python function to count the number of digits in factorial of a given number.

End of Paper